

Built In Functions

1)zip()

In []: `zip()` function combine **from** two **or** more iterables position by position
Syntax: `zip(iterable 1, iterable 2, .....`)

```
In [2]: names=['Yash','Chaitnya','Pratiksha','Prerana','Manjusha','Dhanashree']
marks=[80,85,90,86,84,82]
result=zip(names,marks)
print(list(result))
```

```
[('Yash', 80), ('Chaitnya', 85), ('Pratiksha', 90), ('Prerana', 86), ('Manjusha', 84), ('Dhanashree', 82)]
```

```
In [3]: names=['Yash','Chaitnya','Pratiksha','Prerana','Manjusha','Dhanashree']
marks=[80,85,90,86]
result=zip(names,marks)
print(list(result))
```

```
[('Yash', 80), ('Chaitnya', 85), ('Pratiksha', 90), ('Prerana', 86)]
```

`zip()` function stops when shortest iterable ends.

2)filter()

In []: `filter()` **is** used to select elements that satisfy condition.
Syntax: `filter(function_name, iterable)`

3)map()

In []: `map()` applies a function to every elements of an iterable
Syntax: `map(function_name, iterable)`

4)all()

In []: `all` returns **True** if **all** elements are truthly

```
In [4]: numbers=[10,20,30,40,50]
print(all(numbers))
```

True

```
In [5]: numbers=[10,20,0,40,50]
print(all(numbers))
```

False

```
In [6]: list1=['python',20,30,40,50]
print(all(numbers))
```

False

5)any()

```
In [ ]: any() returns True if at least one elements is truthy.
```

```
In [7]: numbers=[10,20,0,0,50]
print(any(numbers))
```

True

```
In [8]: numbers=[0,0,0,0,0]
print(any(numbers))
```

False

6)chr()

chr() converts a Unicode numbers into a character.
Syntax:chr(number)

```
In [9]: print(chr(65))
```

A

```
In [10]: print(chr(90))
```

Z

```
In [12]: print(chr(97))
```

a

7)ord()

```
In [ ]: ord() does the opposite of chr()
It converts a character into its Unicode number.
```

```
In [13]: print(ord('A'))
```

65

8)bin()

```
In [ ]: bin() function converts an integer into binary representation
```

```
In [14]: print(bin(10))
```

0b1010

```
In [15]: print(bin(5))  
print(bin(8))  
print(bin(15))
```

0b101
0b1000
0b1111

9)format()

```
In [ ]: format() formats a value according to specified format
```

```
In [16]: number=25  
print(format(number, "b"))
```

11001

```
In [17]: number=25  
print(format(number, "d"))
```

25

```
In [18]: number=25  
print(format(number, "x"))
```

19

```
In [19]: price=99.45678  
print(format(price, ".2f"))
```

99.46

10)eval()

```
In [ ]: eval() evaluates a Python expression given as a string
```

```
In [20]: x="10+20"  
result=eval(x)  
print(result)
```

30

```
In [21]: expression="5*10+2"  
print(eval(expression))
```

52

11)abs()

```
In [ ]: abs() returns the absolute value of a number
```

```
In [22]: print(abs(-10))
```

10

```
In [23]: print(abs(10))
```

10

12)round()

```
In [ ]: round() rounds a number to the nearest value or to specified  
number of decimal places.
```

```
In [24]: print(round(10.6))
```

11

```
In [25]: print(round(10.4))
```

10

13)pow()

```
In [ ]: pow() calculates a number raised to a power.  
Syntax:pow(base,exponent)
```

```
In [26]: print(pow(2,3))
```

8

```
In [27]: print(pow(2,4))
```

16

14)divmod()

```
In [ ]: divmod() returns quotient and remainder  
Syntax:divmod(a,b)
```

```
In [28]: result=divmod(17,5)  
print(result)
```

(3, 2)

15)iter()

```
In [ ]: iter() creates an iterator from an iterable
```

```
In [29]: numbers=[10,20,30]  
iterator=iter(numbers)  
print(iterator)
```

<list_iterator object at 0x000001E3CDF683A0>

```
In [30]: print(next(iterator))  
print(next(iterator))  
print(next(iterator))
```

10

20

30

16)next()

```
In [ ]: next() returns the next item from an iterator
```

```
In [32]: list1=[10,20,30,40]  
it=iter(list1)  
print(next(it))  
print(next(it))  
print(next(it))  
print(next(it))
```

10

20

30

40

```
In [ ]: 1)zip()
        2)filter()
        3)map()
        4)all()
        5)any()
        6)chr()
        7)ord()
        8)bin()
        9)format()
        10)eval()
        11)abs()
        12)round()
        13)pow()
        14)divmod()
        15)iter()
        16)next()

        17)min()
        18)max
        19)sum()
        20)print()
        21)reversed()
        22)sorted()
        23)int()
        24)float()
        25)str()
        26)list()
        27)tuple()
        28)set()
        29)dict()
        30)len()
        31)type()
        32)range()
        33)input()
        34)enumerate()
        35)complex()
        36)help()
```